
What Does Repairing Choice Inconsistency Buy?

A Budget-Set Diagnosis

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Repairing an AI agent’s incoherent preferences is an established line of work:
2 inference-time transitivity repairs, isotonic projections onto preference-defined
3 cones, and training-time rationality penalties have all been proposed, several with
4 reported downstream gains. No published result includes a displacement-matched
5 control, so none establishes whether the reported gains come from restoring co-
6 herence or from displacement toward an interior point. We build that control.
7 Given an LLM agent’s own choices over budget sets, we compute the minimal
8 MILP perturbation restoring GARP-consistency, then compare it against a GARP-
9 blind null operator spending the identical displacement budget shrinking every
10 bundle toward a fixed interior point. Under the first payoff, the null outperforms
11 the real repair by more than $2\times$ (mean Δpayoff 0.0220 vs. 0.0091, Wilcoxon
12 $p = 3.98 \times 10^{-10}$, winning 81% of traces). A second payoff built to remove
13 the first one’s exploitable fixed target reproduces the same verdict (null 0.0217
14 vs. real 0.0072, $p = 1.24 \times 10^{-5}$, 70.8% win rate). Trace extremity predicts the
15 null’s advantage in both experiments. The paper’s contributions are a capacity-
16 deconfounded identification strategy for the coherence–competence question and a
17 discard-selection instrument-validity finding. That question remains open.

18 1 Introduction

19 An LLM agent asked to allocate a fixed budget across two goods at varying prices, repeatedly, will
20 frequently violate the Generalized Axiom of Revealed Preference (GARP; Afriat, 1967, Varian,
21 1982): it prefers bundle x_1 to x_2 at one budget line and the reverse at another, in a way no monotone
22 utility function can rationalize. The same budget-allocation instrument is the standard paradigm
23 for testing GARP experimentally on human subjects [Andreoni and Miller, 2002, Choi et al., 2007,
24 2014], and a growing literature applies it to GPT-family models with no repair step, asking whether
25 they satisfy classical rationality axioms [Chen et al., 2023]. A separate literature repairs this kind of
26 inconsistency in LLM systems — restoring transitivity in pairwise judgments, projecting cardinal
27 scores onto monotone cones, and penalizing incoherence during training — and several of these
28 systems report that the repaired outputs are downstream-better than the raw ones. None of them
29 establishes a *dose*: how much better, as a function of how much repair was needed, isolated from the
30 capacity and training confounds that come from comparing a richer model class to a poorer one.

31 An LLM agent makes a sequence of budget-constrained choices over two goods. We compute, as a
32 single mixed-integer linear program, the minimal quantity perturbation that would make the agent’s
33 own realized choice sequence GARP-consistent. The projection is applied post hoc to a frozen agent:
34 nothing about its weights, training, or decoding changes across doses. The size of that perturbation is
35 a graded, cardinal dose. We score both the raw and the repaired sequence against a fixed, equal-weight
36 Cobb–Douglas payoff whose optimum at each budget line is a closed-form function of prices and

37 income alone — never of the agent’s own choices — so the outcome measure is exogenous to the
38 preference data by construction.

39 The design does not identify the effect of coherence restoration. An operator that spends the identical
40 displacement budget shrinking every bundle toward a fixed interior point, with no access to the
41 GARP violation structure, outperforms the real GARP-restoring repair on the exogenous payoff
42 ($p = 3.98 \times 10^{-10}$; §5.1). The result holds at both model scales and under two payoff designs.
43 No published axiom-enforcement result we are aware of includes this control. Restoring GARP-
44 consistency buys no more exogenous payoff than displacement toward the interior of the budget line
45 (0.0091 vs. 0.0220). We label this claim C1; C2 is the coherence–competence question and C3 the
46 claim that the reciprocal-framing manipulation disrupts the agent.

47 The conjunction is new: an agent’s own realized choices, a graded coherence-indexed dose, and an
48 outcome that is not itself a preference judgment, run together so that dose is not confounded with
49 model capacity or training. We run this at a model scale where the pilot found coherence headroom
50 (1.5B parameters), and repeat the same pipeline at 3B, where the pilot found mean baseline CCEI
51 0.99. The 3B run is an identification control on the method. The two scales are not two points on a
52 shared dose curve.

53 The reciprocal-price framing manipulation intended to induce coherence variation at 1.5B produced,
54 in the pilot, a large apparent effect that disappears once discard-selection is corrected with a capped
55 retry protocol: the pilot’s uncorrected estimate and the retry-corrected estimate are both statistically
56 indistinguishable from zero. The discard rate is the more reliable signal that the manipulation
57 disrupted the agent (our claim C3): it is measured without conditioning on completion, and the CCEI
58 point estimate is not. The gap between the uncorrected and the corrected estimate quantifies how
59 much a naive discard rule can distort a measured effect (§5.2). Replacing single-turn elicitation with
60 25 sequential calls at 1.5B drops the GARP pass rate from 0.40 to 0.10, with no discard confound in
61 either condition. The drop runs opposite to an independently published finding in the same domain
62 and model family at a larger scale [Wang et al., 2025a].

63 **Contributions.** (1) An application of Demuyneck and Rehbeck [2023]’s minimal-quantity-error MILP
64 projection to an agent’s own realized choices, verified independently on every output, not by solver
65 status. (2) A closed-form exogenous payoff, fixed before any data collection, checked for leakage and
66 for the mechanical confound that larger violations have more room to improve. (3) A dose-matched,
67 GARP-blind null-operator control, run against two exogenous payoff designs. The measured raw
68 dose–response relationship does not show that restoring GARP-consistency buys the exogenous
69 payoff gain: displacement toward the interior buys as much (§5.1). (4) An instrument-validity finding
70 (discard-selection bias under a disruptive manipulation), corrected by a capped retry protocol and
71 reported per cell. (5) A single-turn-vs-multi-turn elicitation-format manipulation that drops the
72 GARP pass rate from 0.40 to 0.10 at 1.5B. The drop runs opposite to an independently published
73 single-turn-vs-multi-turn finding in the same budget-allocation domain and the same model family
74 (Qwen2.5) at a larger scale [Wang et al., 2025a].

75 2 Related Work

76 Four criteria jointly define the claim: an agent’s own choices, an exogenous payoff, a graded rather
77 than binary dose, and a minimum-distance projection. Of nine published repair systems, none satisfies
78 all four (Table 1).

79 **POISE.** Wang et al. [2026] project with pool-adjacent-violators onto a closed convex chain-monotone
80 cone, and prove the edit lands weakly closer to a posited ground truth. The GARP-consistent set
81 is a union of polyhedra, hence not convex, so the non-expansiveness of projection onto a closed
82 convex set in L_2 does not transfer (§B.1); the minimum-distance priority is theirs. POISE projects
83 a *teacher’s* offline labels. Three further inference-time repairs have a binary edit and no Afriat
84 machinery: Chadwick et al. [2025], TrustJudge [Wang et al., 2025b], and CONSISTRE [Sun, 2026].
85 LLM-RankFusion [Zeng et al., 2024] has an exogenous payoff but declines a minimum-distance
86 method. Buchanan and Foster [2026] alters the agent itself at training time and reports no downstream
87 comparison. Aguiar and Kashaev [2026] is not an LLM agent and is evaluated on forecasting.

88 **Preference-model variants.** A parallel line, outside the table because it varies the transitivity of the
89 *preference model* rather than an agent’s choices, makes the cycle-tolerant arm a strict superset model

Table 1: Nine published repair systems scored against the four criteria that define the claim. *Own choices*: the repaired object is a choice the agent made for itself from a budget set. *Exogenous payoff*: an outcome into which no preference judgment enters; *graded dose*: the intervention has a continuous magnitude, not an on/off setting; *min.-distance projection*: the edit is the solution to an explicit minimum-distance program.

	Own choices	Exogenous payoff	Graded dose	Min.-distance projection
POISE [Wang et al., 2026]	no	no	no	yes
Rationality layer [Chadwick et al., 2025]	no	no	no	yes
TrustJudge [Wang et al., 2025b]	no	no	no	no
CONSISTRE [Sun, 2026]	no	partial	no	no
LLM-RankFusion [Zeng et al., 2024]	no	yes	no	no (declined)
Innate preferences [Buchanan and Foster, 2026]	yes	no	no	no
GARP-EFM [Aguar and Kashaev, 2026]	no	yes	no	no
HRC/DSPPO [Huang et al., 2026]	no	no	yes	no
Control vectors [Cook et al., 2026]	yes	no	yes	no
This paper	yes	yes	yes	yes

class: GPM [Zhang et al., 2025] and HRC/DSPPO [Huang et al., 2026]. Coherence is therefore confounded with capacity by construction, and on GPM’s own length-controlled metric that arm loses in 18 of 24 cells. HRC/DSPPO’s inverted-U dose–response curve is over a training-time schedule weight on a learned proxy, scored by an LLM judge. Ours acts post hoc on a payoff that is exogenous by construction. Andrews [2026] proposes 1 – CCEI as a training penalty on a priori grounds and reports no measurement of it. We report the measurement that proposal calls for.

Psychometric reliability. Nitsch et al. [2022] find that CCEI and Houtman–Maks [Houtman and Maks, 1985] reliability never reaches acceptable levels across eight datasets, and that mean CCEI fell among participants given a chance to revise their own choices. Their revision arm re-presented a random subset of trials with no consistency objective, so it could not guarantee fewer violations; ours attains rationalizability by construction. Repair by construction rather than by re-presentation also separates ours from Yamin et al. [2026]’s isotonic repair, which their own evaluation reports as worse in 14 of 16 cells. Nitsch et al. diagnose their finding as a between-subject-variance problem whose prescribed remedy, a between-groups design, is this paper’s design (§6 bounds how far our power supports it). Three independently published axiom-enforcement results — Nitsch et al. [2022], Yamin et al. [2026], and Zhu et al. [2025] — all report that enforcing axioms failed to improve the outcome measured. We therefore carry a null-operator control that none of the three had.

Economic-choice steering. Cook et al. [2026], in the economics literature, steer economic choices toward payoff-maximizing behavior with personas and control vectors. They meet two of the four criteria and use no Afriat machinery. They find that reframing moves models while persona prompting does not, which corroborates our manipulation choice. Wang et al. [2025a] find the same in the same domain and model family (Qwen2.5) at 7B, where collapsing multi-turn to single-turn drops CCEI by 0.241. Which format is more degrading reverses between their 7B model and ours: reversing their manipulation reverses the sign, and the GARP pass rate falls from 0.40 to 0.10 (§5.3).

3 Method

3.1 Minimal-quantity-error MILP projection

The MILP formulation below is Demuynck and Rehbeck [2023]’s multiplier-free ordinal characterization of minimal-quantity-error GARP repair, applied here to an LLM agent’s own choice sequence.

For a sequence of T observations $(p_t, x_t)_{t=1}^T$, K goods, GARP holds if there is no sequence of choices such that x_{t_1} is revealed preferred to x_{t_2} , $\dots$, x_{t_n} is revealed preferred to x_{t_1} , with at least one strict revealed preference in the cycle. Given a GARP-violating sequence, we seek the minimal perturbation $\tilde{x}_t$ that restores GARP-consistency. The Afriat-multiplier formulation is bilinear in the unknowns once $\tilde{x}$ is itself a decision variable; we instead use the multiplier-free ordinal characterization of Demuynck and Rehbeck [2023], under which every constraint is linear. At $T = 25$ and $K = 2$

this is 600 binaries and 125 continuous variables per trace, solved as a single MILP on the HiGHS backend [Huangfu and Hall, 2018]. The full formulation and the three method commitments are in Appendix B. All 85 projections were re-verified GARP-consistent by the Warshall-closure GARP check [Warshall, 1962], independently of the solver (MIP gap $\leq 8.1 \times 10^{-5}$ on every solve).

The objective is L_1 : minimal total absolute displacement between $\tilde{x}$ and x across goods and rounds (L_∞ is recorded alongside; L_2 would require an MIQP formulation). Weights are uniform, which keeps the program a pure MILP. The *dose* for a trace is its L_1 projection distance $\sum_t \sum_k |x_{t,k} - \tilde{x}_{t,k}|$ (Figure 1, step 5), comparable across sessions without further normalization because income is fixed throughout. It is the same style of graded, minimum-cost severity measure as Dean and Martin [2016], applied here to an LLM’s own choices and paired with the exogenous payoff of §3.2. Wherever a mean dose or mean Δpayoff is reported over a group of traces, it is averaged over every trace in that group, with 0 for any already-GARP-consistent trace. We compute a Cobb–Douglas demand whose expenditure shares are fitted to each trace’s observed data as an upper bound on the reported distance; it is not supplied to the solver (Appendix A, check 1).

Guarantees. Because the GARP-consistent set is a union of polyhedra rather than a single convex set, we claim no distance-minimization guarantee analogous to a convex-cone projection [Wang et al., 2026].

3.2 The exogenous payoff

The payoff is fixed independently of the agent’s revealed choices. A utility function fit to the agent’s choices (such as the share-fitted Cobb–Douglas demand of §3.1) would be circular if used as the outcome measure, rewarding the agent for resembling its own revealed preferences. We use an equal-weight Cobb–Douglas valuation $U_{\text{exo}}(x) = x_A^{0.5} x_B^{0.5}$, fixed before any model was queried. It was never re-estimated from any agent’s choices. At budget line (p_t, I_t) the optimal bundle x_t^* is the closed-form Cobb–Douglas demand $x_{t,A}^* = 0.5I_t/p_{t,A}$, $x_{t,B}^* = 0.5I_t/p_{t,B}$, which depends only on prices and income. The payoff score is the efficiency ratio $\text{payoff}(x_t) = U_{\text{exo}}(x_t)/U_{\text{exo}}(x_t^*) \in (0, 1]$, computed identically for the raw and projected sequence at the same budget lines, so that $\Delta\text{payoff} = \text{payoff}(\tilde{x}) - \text{payoff}(x)$ is comparable across conditions, models, and repairs. This has the structure of a money-metric utility index, the standard welfare-comparison device over the same GARP/Afriat objects this paper’s method already relies on. Ouyang et al. [2024] give one of the few comparable exogenous, real-world payoffs: they score LLM-generated forecasts against firms’ realized future spending and find a *non-monotonic* alignment-accuracy relationship. Separating an alignment-specific effect from a generic fine-tuning effect requires a placebo of matched size, which that design lacks. Our design makes that separation.

We checked four failure modes of this design: a shared-data leak between payoff and projection, a mechanical link between GARP-consistency and payoff, a projection direction correlated with the payoff optimum, and the confound that larger-dose traces start with lower payoff and have more room to improve. All four are ruled out — the fourth, the strongest of them, leaves the dose coefficient significant once raw payoff is controlled for (Appendix A gives every check and number) — so the raw dose–response statistic of §5.1 is not attributable to any of them. A fifth concern, that this payoff’s concavity could reward displacement alone, motivates the control constructed next.

3.3 Null-operator controls

The payoff of §3.2 reduces to $\text{payoff}(s) = 2\sqrt{s(1-s)}$, where s is the expenditure share on good A. It is concave with a single interior maximum at $s = 0.5$, identical across every trace. Any operator that moves a bundle toward that fixed point raises that bundle’s payoff whether or not it restores GARP-consistency. We construct a *null operator* that shrinks every observed bundle in a GARP-violating trace toward the exogenous optimum x_t^* by a single shared factor $\lambda \in [0, 1]$, chosen so the null’s total L_1 displacement exactly matches the real projection’s dose. The null does not call the GARP check and is not guaranteed to restore consistency; it is scored by the payoff of §3.2.

Because that fixed target is identical across every trace, we build a second payoff that removes it: a per-trace Cobb–Douglas weight $\alpha_s \sim \text{Uniform}(0.05, 0.95)$, drawn independently for each of the 85 traces and each of $K = 20$ replicate draws, read only at scoring time. We report two null constructions under this payoff: a *primary* null, unaware of α_s , and an *oracle* null that aims at the

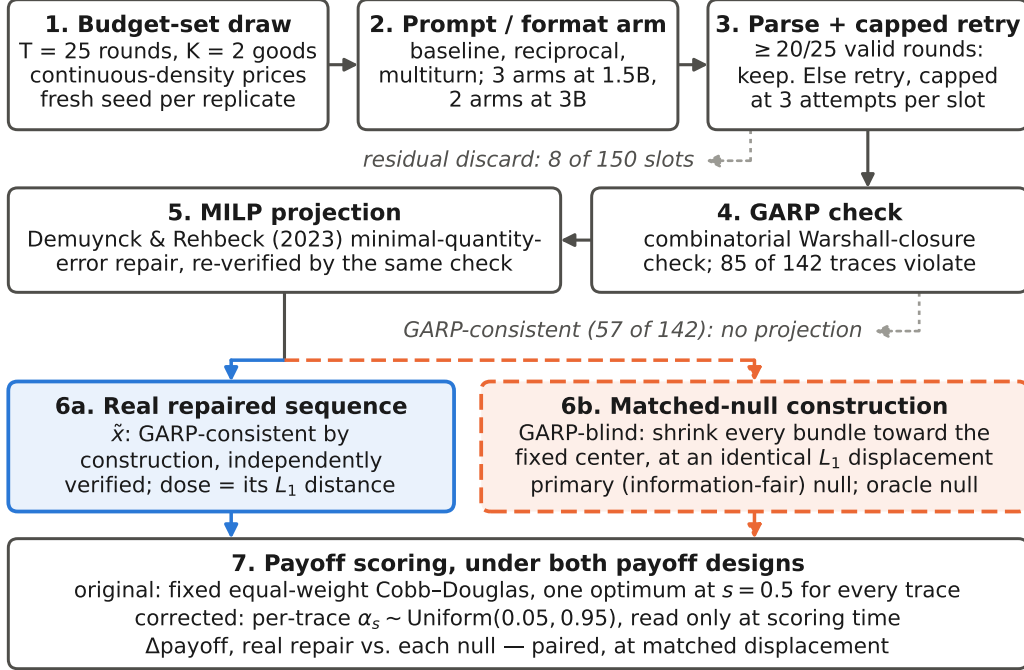


Figure 1: The pipeline, one replicate end to end: budget-set draw (1); prompt/format condition (2); parse (3); GARP check (4); and, for the traces that violate it, the minimal-quantity-error MILP of §3.1 (5). The real repaired sequence (6a, solid blue) and the GARP-blind null spending an identical L_1 displacement (6b, dashed orange) are scored (7) under both payoff designs.

177 true per-trace optimum $x_{\alpha_s}^*$. The two are reported separately. The oracle null is an upper bound: it
 178 uses the per-trace α_s , which a repair algorithm does not observe.

179 4 Experimental design

180 4.1 Models and conditions

181 We use two locally hosted open-weight models: `qwen2.5:1.5b-instruct` at 1.5B and
 182 `llama3.2:3b-instruct` at 3B. The pilot found within-condition coherence variation at 1.5B and
 183 mean baseline CCEI 0.99 at 3B, so 1.5B carries the dose-response measurement and 3B serves as a
 184 low-headroom control. Models were run one at a time, all runs for one model completing before the
 185 next was loaded; concurrent residency of a third model family caused reproducible crashes. **Compute.**
 186 All inference ran locally on 4-bit-quantized weights, with no GPU and no commercial API calls. The
 187 full $N = 30$ design completed in ≈ 2.1 hours wall-clock, every MILP projection (§3.1) solving in
 188 under 5 seconds.

189 Three conditions at 1.5B, two at 3B. **Baseline.** Direct per-token-return framing, single turn, all 25
 190 rounds in one prompt and response. **Reciprocal.** The same single-turn format with an inverted-price
 191 framing manipulation, the strongest manipulation identified in the pilot. **Multi-turn** (1.5B only).
 192 Baseline framing, but each of the 25 rounds is a separate sequential call carrying the running conver-
 193 sation history, which isolates the elicitation-format mechanism from the price-framing mechanism.
 194 Wang et al. [2025a] collapse multi-turn to single-turn in the same domain and find that this drops
 195 CCEI by 0.241 for a same-family model at 7B; because our baseline is already single-turn, this
 196 condition tests the reverse. The 3B model is a low-headroom control, so this condition was run at
 197 1.5B only.

4.2 Budget sets and power

Each of $N = 30$ independent replicates per (model, condition) cell draws a fresh $T = 25$, $K = 2$ budget set (prices $p_A, p_B \sim U[0.1, 1.0]$ i.i.d., rejected unless $\max(p_A, p_B) \geq 0.5$, budget exhaustion enforced), seeded independently. Independent seeding supports between-condition replication. A slot is one such design position; each slot is run for up to three attempts, one session each (§4.3); a completed attempt yields one trace. Continuous, independently drawn prices with enforced budget exhaustion also avoid a pathology documented by Andrews [2026]: CCEI can read exactly 1.0 on GARP-violating data whenever a comparison lands on exact budget-line equality, common under grid-sampled or rounded prices. $N = 30$ gives power ≥ 0.999 against the pilot’s full framing-effect magnitude on the pass-rate metric and ≥ 0.80 down to $\sim 60\%$ attenuation, but is underpowered for a CCEI-scale effect of the pilot’s magnitude at 1.5B (minimum detectable ≈ 0.11). Every replicate’s Bronars power of the budget set [Bronars, 1987] is recorded; all 150 draws held Bronars power ≥ 0.998 .

4.3 Discard selection and the retry protocol

The pilot found that under reciprocal framing at 1.5B, 52% of sessions (13 of 25) failed the required output format and were silently discarded under the pilot’s handling. Silent discarding is standard practice; this paper measures its consequences. This is a claim about instrument validity. The main experiment implements a capped retry protocol: a session failing the $\geq 20/25$ -valid-rounds threshold is retried up to three times with a new random seed; a slot still failing after three attempts is a residual discard, excluded from the main CCEI/GARP analysis but scored separately for the audit in Table 2. Three of the 30 qwen reciprocal slots returned zero valid rounds on all three attempts, yielding no price or quantity data.

At 1.5B, retry-rescued sessions differ from first-attempt sessions: mean CCEI 0.9651 against 0.9315, but GARP pass rate 0.1429 against 0.4706. The residual-discard group has the lowest mean CCEI of the three (0.8821), though its GARP pass rate is measured on only 3 slots. The residual sample is too small ($n = 6$, 3 measurable) to be conclusive, but the pattern is consistent with the pilot’s selection concern surviving the retry correction at reduced magnitude. At 3B, the first-attempt and retry-rescued groups differ by 0.0118 in CCEI and 0.143 in GARP pass rate, so the selection concern shows up more sharply at 1.5B than at 3B.

Table 2: Discard-outcome breakdown for all 60 reciprocal-framing sessions (30 per model), by attempt group: kept on the first attempt, rescued by a later retry, or still discarded after three attempts (§4.3). GARP pass, CCEI and mean dose (L_1) follow the conventions of §3.1 and Table 3. *Computed over the 3 of 6 qwen residual-discard slots with any valid rounds.

Model	Attempt group	n	mean CCEI	GARP pass	mean dose (L_1)
llama3.2:3b	first-attempt success	21	0.9742	0.4286	5.086
llama3.2:3b	retry-rescued	7	0.9624	0.2857	9.839
llama3.2:3b	residual discard	2	1.0000	1.0000	0.000
qwen2.5:1.5b	first-attempt success	17	0.9315	0.4706	13.847
qwen2.5:1.5b	retry-rescued	7	0.9651	0.1429	7.600
qwen2.5:1.5b	residual discard	6	0.8821*	0.6667*	24.196*

5 Results

We ran 150 slots (llama3.2:3b in 2 conditions, qwen2.5:1.5b in 3, 30 replicates each) and made 187 attempts. Of these, 142 slots kept a usable trace and 8 were residual discards after three attempts. Every cell was run to its full 30 slots.

5.1 Dose-response and the null-operator control

Across the 85 GARP-violating traces (60 at 1.5B, 25 at 3B): Spearman $\rho = 0.729$ ($p < 0.0001$), Pearson $r = 0.821$ ($p < 0.0001$), mean $\Delta\text{payoff} +0.0091$ (one-sample $t = 5.41$, $p < 0.00001$), with 70 of 85 traces (82%) improving after repair and 15 (18%) worsening. The relationship holds

Table 3: Headline results per (model, condition) cell. GARP pass and CCEI are computed on kept traces only; dose and Δpayoff are averaged over all kept traces, with 0 for any already-GARP-consistent trace, and are under the original payoff (§3.2).

Model	Condition	n kept	mean CCEI	GARP pass	mean dose (L_1)	mean Δpayoff
llama3.2:3b	baseline	30	0.9900	0.73	5.01	+0.0018
llama3.2:3b	reciprocal	28	0.9713	0.39	6.27	+0.0016
qwen2.5:1.5b	baseline	30	0.9522	0.40	16.68	+0.0090
qwen2.5:1.5b	multi-turn	30	0.9540	0.10	14.54	+0.0083
qwen2.5:1.5b	reciprocal	24	0.9413	0.38	12.02	+0.0065

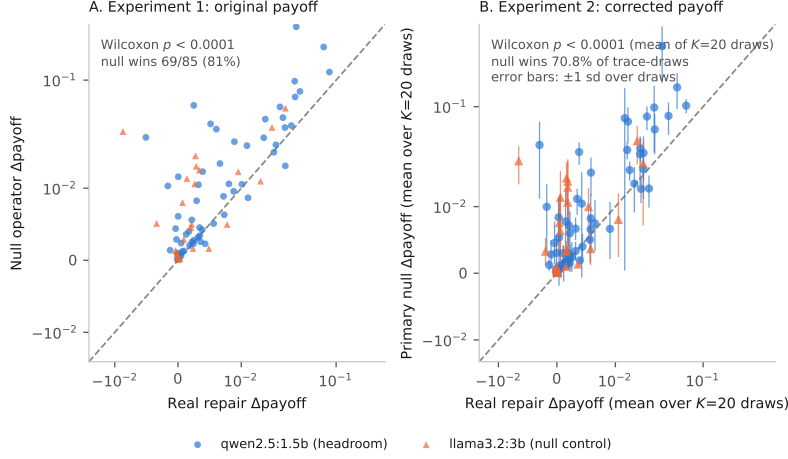


Figure 2: Real repair’s Δpayoff (x-axis) against the dose-matched null operator’s Δpayoff (y-axis) on symmetric-log axes, one point per GARP-violating trace ($n = 85$; circle/blue: qwen2.5:1.5b, triangle/orange: llama3.2:3b). Dashed line is $y = x$; points above it have higher null Δpayoff than real. (A) Experiment 1, original payoff; (B) Experiment 2, corrected payoff, with each trace’s primary-null Δpayoff averaged over $K = 20$ draws of the random target (± 1 SD error bars).

at each scale separately: $\rho = 0.756$ ($p < 0.0001$, $n = 60$) at 1.5B and $\rho = 0.614$ ($p = 0.0011$, $n = 25$) at 3B. We do not compare its magnitude across scale. We tested the mechanical confound that larger-dose traces might start from worse raw payoff and therefore have more room to improve. The confound is real (dose vs. raw payoff, Pearson $r = -0.41$, $p < 0.0001$) but does not explain the relationship away. The partial correlation of dose against Δpayoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$). An OLS regression of Δpayoff on dose and raw payoff jointly finds dose significant ($t = 11.4$, $p < 10^{-16}$), with raw payoff’s marginal contribution not distinguishable from zero ($p = 0.18$) once dose is included (Appendix A).

The raw-payoff confound bears on the *severity* of the raw violation. It leaves open whether restoring GARP-consistency or the displacement does the work. The null operator (§3.3) spends the identical L_1 budget as the real repair and shrinks every bundle toward the exogenous payoff’s fixed optimum, using no GARP information.

Experiment 1 (original payoff). The null operator outperforms the real repair by $2.4\times$: mean $\Delta\text{payoff}_{\text{null}} = 0.0220$ vs. mean $\Delta\text{payoff}_{\text{real}} = 0.0091$, Wilcoxon signed-rank $p = 3.98 \times 10^{-10}$, with the null winning 69 of 85 traces (81%; Fig. 2A). This holds at both model scales. The partial correlation of dose against $\Delta\text{payoff}_{\text{real}}$, controlling for the null’s mechanically predicted gain, remains positive and significant ($r = 0.574$, $p = 1.13 \times 10^{-8}$), so dose retains information beyond what the null predicts.

Experiment 2 (corrected payoff). The original payoff’s exploit is a single fixed target ($s = 0.5$) identical across every trace. Experiment 2 draws an independent Cobb–Douglas weight $\alpha_s \sim \text{Uniform}(0.05, 0.95)$ per trace, over $K = 20$ independent draws (§3.3). Experiment 2 reproduces Experiment 1: the primary null still outperforms the real repair, by $3.0\times$ (mean $\Delta\text{payoff}_{\text{null}} =$

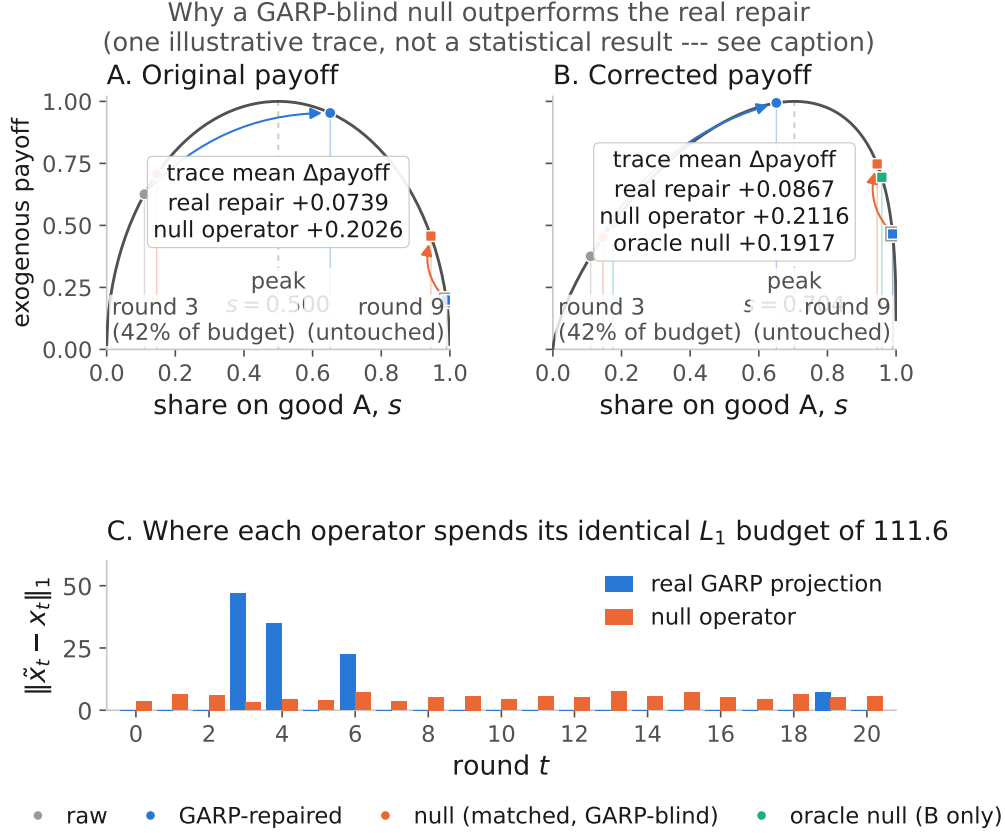


Figure 3: The largest-dose trace (qwen2.5:1.5b, reciprocal, replicate 12; $L_1 = 111.64$, $T = 21$ rounds). With $K = 2$, payoff is a function of expenditure share s . **(A)** The original payoff $2\sqrt{s(1-s)}$, peak at $s = 0.5$; **(B)** the corrected payoff at this trace’s draw $\alpha_s = 0.704$; **(C)** per-round s across all 21 rounds for the raw trace, the real repair, and the null operator.

0.02167 vs. mean $\Delta\text{payoff}_{\text{real}} = 0.00723$, Wilcoxon $p = 1.24 \times 10^{-5}$, win rate 70.8%), significant in 20 of 20 draws. The oracle null, privileged to know the true per-trace target, wins 80.4% of traces (mean $\Delta\text{payoff}_{\text{null}} = 0.02224$, $p = 1.35 \times 10^{-9}$, significant in 20 of 20 draws). The partial correlation of dose with $\Delta\text{payoff}_{\text{real}}$, controlling for the null, attenuates from Experiment 1’s $r = 0.574$ to a mean $r = 0.37$ across the 20 draws, significant in 14–15 of 20.

Two payoff designs. One is the paper’s original; the second was built to remove the first one’s exploitable structure. Both return the same negative paired comparison. A third payoff, selected after seeing that the second was also negative, would be payoff-shopping (§6).

Trace extremity and the null’s advantage. Trace extremity ($1 - \text{raw payoff}$ under the original fixed payoff, a measure of the raw bundles independent of which payoff later scores them) predicts the null’s advantage in both experiments (Experiment 1: Pearson $r = 0.679$; Experiment 2 primary null: mean $r = 0.631$ across 20 draws), and predicts the real repair’s own gain less strongly in both. This pattern does not separate “centering” from a more general account on which moving an extreme trace toward the interior helps regardless of destination, since Experiment 2’s primary null is mechanically the same fixed-center operator as Experiment 1’s. Experiment 2’s oracle null, which targets a per-trace-varying point, shows the same extremity-advantage to within 0.01 (mean $r = 0.624$ against 0.631). We treat the “escape a bad start” reading as a hypothesis for the discussion; we do not test it. Figure 3 illustrates it on a single trace. On the largest-dose trace the repair changes 5 of 21 rounds, spending 42% of its L_1 budget on round 3 alone (s 0.11 $\rightarrow$ 0.65, payoff 0.626 $\rightarrow$ 0.953) and leaving the other 16 at a near-corner $s = 0.99$, while the null spreads the same total displacement over all 21 rounds.

The hypothesis that restoring GARP-consistency, beyond displacement toward the interior of the budget line, buys more exogenous payoff (C1) is not supported under either payoff design. The raw dose–response relationship holds; it is not evidence for the specific mechanism C1 claims.

5.2 Reciprocal framing and discard selection

At 3B, reciprocal framing replicates the pilot’s finding on an independently drawn budget set: GARP pass rate drops from 0.73 (22/30) to 0.39 (11/28), $p = 0.0089$ (Benjamini–Hochberg $p_{BH} = 0.011$, Holm $p_{Holm} = 0.062$); CCEI does not reach significance (0.9900 vs. 0.9713, t -test $p = 0.0508$). CCEI compresses near 1, as the pilot found. Here, GARP pass rate separated baseline from reciprocal and CCEI did not.

At 1.5B, the reciprocal-framing effect does not survive correction for discard-selection. First-attempt discard under reciprocal framing was 43.3% (13/30), against the pilot’s unretried 52%, falling to a residual 20.0% (6/30) after the retry protocol. Reciprocal framing produces more discards than any other condition, including the other 1.5B manipulation (multi-turn, 0%). Once the surviving 24 sessions are compared, GARP pass rate (0.40 vs. 0.38, $p = 0.85$) and CCEI (0.9522 vs. 0.9413, $p = 0.73$) are indistinguishable between baseline and reciprocal framing. Neither experiment detects a CCEI shift: the pilot’s +0.0169 ($p = 0.66$) and the main experiment’s -0.0109 ($p = 0.73$) are two statistically indistinguishable nulls. Table 2 breaks the discards down by attempt.

5.3 Multi-turn elicitation format

Replacing the single prompt with 25 sequential calls drops the GARP pass rate from 40% to 10% ($p = 0.0073$) while CCEI does not move (0.9522 vs. 0.9540, $p = 0.91$; Table 4), with zero discards in either condition. Reciprocal framing produced no detectable drop at 1.5B, so the multi-turn format is the only 1.5B manipulation with a measured pass-rate effect. The drop survives Benjamini–Hochberg correction ($p_{BH} = 0.0094$) but not Holm ($p_{Holm} = 0.058$). Together with the 3B reciprocal-framing result (§5.2), GARP pass rate again separates the conditions and CCEI does not. Wang et al. [2025a] report the opposite: in the same domain, on the same model family at 7B (Qwen2.5-7B), single-turn elicitation was more coherence-degrading than multi-turn; here, at 1.5B, the reverse holds. We have no explanation for the reversal.

Table 4: 1.5B (qwen2.5:1.5b), baseline vs. multi-turn elicitation format (25 sequential calls), $n = 30$ per condition. GARP pass rate is reported as count/ n with 95% Wilson score confidence intervals (Pearson χ^2 test, $p = 0.0073$); mean CCEI is reported with 95% t -based confidence intervals ($df = 29$; two-sample t -test, $p = 0.91$).

Condition	GARP pass	95% CI (Wilson)	mean CCEI [95% CI (t)]
baseline	12/30 (40.0%)	[24.6, 57.7]%	0.9522 [0.9253, 0.9791]
multi-turn (25 seq. calls)	3/30 (10.0%)	[3.5, 25.6]%	0.9540 [0.9350, 0.9729]

6 Limitations

CCEI is underpowered at 1.5B. The 1.5B model’s within-condition CCEI noise (baseline WSCV 14.8%) is comparable to published human test–retest noise on the same index [Nitsch et al., 2022], and $N = 30$ is underpowered for a CCEI-scale effect of the pilot’s magnitude: the minimum detectable effect is $2.5\times$ the pilot’s estimate. A CCEI-power-matched design would need $N = 111$ – 161 ; we did not run that design. The pass-rate metric is powered ≥ 0.999 throughout, and every CCEI result carries its own confidence interval.

Single run per condition. Every (model, condition) cell is one run of the stated protocol. There are no repeated full-pipeline runs, so we report no between-run variance on the headline pass-rate and Δ payoff statistics, only the within-run replicate variance in the reported confidence intervals and p -values.

One scale point per model. The two-model design isolates the identification strategy from a scale confound but does not trace a dose–response relationship *across* scale. In our model-selection sweep

(§4), coherence headroom and measurement reliability did not coexist in any model tested, and a third model family was excluded because it could not be run alongside the others.

Two exogenous payoffs. We report two payoff designs (§3.2, §3.3). The first payoff’s single, globally identical optimum is exploitable by a GARP-blind operator; the second removes that property and reproduces the same verdict. A third payoff, selected after seeing that the second was also negative, would be payoff-shopping, which is the researcher-degree-of-freedom problem an exogenous payoff was introduced to rule out (§5.1).

Consistent with three prior negative results. Three independently published repair attempts moved the wrong way (§2). The null-operator result of §5.1 is a fourth: under two payoffs, restoring GARP-consistency does not outperform a dose-matched GARP-blind operator. Separately, our reciprocal-framing manipulation at 1.5B (§5.2) failed to induce the coherence variation it was designed to induce. Net of displacement, this paper’s attempt to find a positive coherence–competence relationship did not succeed. That leaves C2 open; it is not evidence that the relationship is negative in general.

7 Broader Impacts

This paper’s result argues against imposing GARP repair on AI economic agents as a routine intervention applied without a matched control: under two payoff designs, a GARP-blind operator that displaces choices toward an interior point outperforms the real repair (§5.1). A practitioner deploying coherence repair on the strength of a raw within-scale dose–response correlation alone would be acting on the confounded estimate this paper’s control was built to separate. Correcting the discard selection removes the reciprocal-framing manipulation’s apparent effect on the coherence it was designed to disrupt (§5.2). The design measures whether repair helps. Here it does not. The discard-selection finding (§4.3) is not specific to our instrument: a consistency instrument that drops disruption-caused failures without recording them will underestimate the disrupting manipulation. We see no elevated misuse risk specific to this work: synthetic tasks, open-weight models, no new dataset, no human subjects.

8 Conclusion

Restoring GARP-consistency in an LLM agent’s own realized choices does not buy more exogenous payoff than displacement alone. Applying the minimal-quantity-error GARP repair of Demuyneck and Rehbeck [2023] post hoc to a frozen agent, we find a monotone raw within-scale dose–response relationship between dose and Δpayoff . At both model scales and under two payoff designs, a dose-matched, GARP-blind null operator outperforms the real repair on that same payoff ($p = 3.98 \times 10^{-10}$ and $p = 1.24 \times 10^{-5}$). C1 is not supported.

Trace extremity predicts the null’s advantage in both experiments, which offers a candidate explanation for that advantage but does not establish it. The paper’s identification argument (C2) survives: the coherence–competence relationship remains open, and this is the first attempt we know of to measure it without confounding coherence with capacity, a preference-judgment outcome, or displacement magnitude. A practitioner evaluating coherence repair should include a dose-matched null operator before reading a dose–response correlation as evidence of benefit.

Two further findings stand on their own. A framing manipulation’s apparent effect was a discard-selection artifact (C3). An elicitation-format manipulation moved GARP pass rate in the direction opposite to that reported by Wang et al. [2025a], in the same domain and model family at a larger scale.

References

- S. N. Afriat. The construction of utility functions from expenditure data. *International Economic Review*, 8(1):67–77, 1967.
- Victor H. Aguiar and Nail Kashaev. GARP-EFM: Improving foundation models with revealed preference structure. *arXiv preprint arXiv:2603.23993*, 2026.
- James Andreoni and John Miller. Giving according to GARP: An experimental test of the consistency of preferences for altruism. *Econometrica*, 70(2):737–753, 2002.

367 Isaiah Andrews. Revealed rationality: Label-free evaluation and regularization from representation
368 theorems. *arXiv preprint arXiv:2608.05015*, 2026.

369 Stephen G. Bronars. The power of nonparametric tests of preference maximization. *Econometrica*,
370 55(3):693–698, 1987.

371 Joy Buchanan and Joshua Foster. The innate economic preferences of language models. *arXiv*
372 *preprint arXiv:2607.26288*, 2026.

373 Alina Chadwick, Anson Kahng, and Jens Kipper. Dutch books and money pumps: rectifying
374 vulnerabilities in LLMs through rationality. In *Proceedings of the 4th International Conference on*
375 *Human and Artificial Rationality (HAR)*, 2025.

376 Yiting Chen, Tracy Xiao Liu, You Shan, and Songfa Zhong. The emergence of economic rationality
377 of GPT. *Proceedings of the National Academy of Sciences*, 120(51):e2316205120, 2023.

378 Syngjoo Choi, Raymond Fisman, Douglas Gale, and Shachar Kariv. Consistency and heterogeneity
379 of individual behavior under uncertainty. *American Economic Review*, 97(5):1921–1938, 2007.

380 Syngjoo Choi, Shachar Kariv, Wieland Müller, and Dan Silverman. Who is (more) rational? *American*
381 *Economic Review*, 104(6):1518–1550, 2014.

382 Thomas R. Cook, Sophia Kazinnik, Zach Modig, and Nathan M. Palmer. What do LLMs want?
383 Technical Report FEDS 2026-006, Federal Reserve Board, 2026.

384 Mark Dean and Daniel Martin. Measuring rationality with the minimum cost of revealed preference
385 violations. *Review of Economics and Statistics*, 98(3):524–534, 2016.

386 Thomas Demuynck and John Rehbeck. Computing revealed preference goodness-of-fit measures
387 with integer programming. *Economic Theory*, 76(4):1175–1195, 2023.

388 M. Houtman and J. A. Maks. Determining all maximal data subsets consistent with revealed
389 preference. *Kwantitatieve Methoden*, 19:89–104, 1985.

390 Huang, Li, Zhao, and Li. Transitivity meets cyclicity: Explicit preference decomposition for dynamic
391 llm alignment. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2605.17342.

392 Q. Huangfu and J. A. J. Hall. Parallelizing the dual revised simplex method. *Mathematical Program-*
393 *ming Computation*, 10(1):119–142, 2018.

394 Felix J. Nitsch, Luca M. Lüpken, Nils Lüscho, and Tobias Kalenscher. On the reliability of
395 individual economic rationality measurements. *Proceedings of the National Academy of Sciences*,
396 119(31):e2202070119, 2022.

397 Shumiao Ouyang, Hayong Yun, and Xingjian Zheng. AI as decision-maker: Ethics and risk prefer-
398 ences of LLMs. *arXiv preprint arXiv:2406.01168*, 2024.

399 Mingxuan Sun. CONSISTRE: A unified consistency-aware framework for document-level relation
400 extraction with large language models. *arXiv preprint arXiv:2607.24312*, 2026.

401 Hal R. Varian. The nonparametric approach to demand analysis. *Econometrica*, 50(4):945–973,
402 1982.

403 Wang, Yao, Zhang, Gai, Liu, and Zhong. When experimental economics meets large language
404 models: Tactics with evidence. *arXiv preprint arXiv:2505.21371*, 2025a.

405 Wang, Zhan, et al. Trustroboreward: Preference-ordered isotonic score editing for multi-paradigm
406 robot reward modeling. *arXiv preprint arXiv:2608.08491*, 2026.

407 Yidong Wang, Yunze Song, Tingyuan Zhu, Xuanwang Zhang, Zhuohao Yu, Hao Chen, Chiyu
408 Song, Qiufeng Wang, Cunxiang Wang, Zhen Wu, Xinyu Dai, Yue Zhang, Wei Ye, and Shikun
409 Zhang. Trustjudge: Inconsistencies of LLM-as-a-judge and how to alleviate them. *arXiv preprint*
410 *arXiv:2509.21117*, 2025b.

411 Stephen Warshall. A theorem on Boolean matrices. *Journal of the ACM*, 9(1):11–12, 1962.

- 412 Kaiser Yamin, J. Tang, S. Cortes-Gomez, A. Sharma, Eric Horvitz, and Bryan Wilder. When
 413 agents say one thing and do another: Validating elicited beliefs from LLMs. *arXiv preprint*
 414 *arXiv:2602.06286*, 2026.
- 415 Yifan Zeng, Ojas Tendolkar, Raymond Baartmans, Qingyun Wu, Lizhong Chen, and Huazheng
 416 Wang. LLM-RankFusion: Mitigating intrinsic inconsistency in LLM-based ranking. *arXiv preprint*
 417 *arXiv:2406.00231*, 2024.
- 418 Zhang, Zhang, Wu, Xu, and Gu. Beyond bradley-terry models: A general preference model for
 419 language model alignment. In *International Conference on Machine Learning (ICML)*, 2025.
 420 *arXiv:2410.02197*.
- 421 Jian-Qiao Zhu, Haijiang Yan, and Thomas L. Griffiths. Recovering event probabilities from large
 422 language model embeddings via axiomatic constraints. *arXiv preprint arXiv:2505.07883*, 2025.

423 A Payoff exogeneity audit

- 424 **(1) Payoff inputs.** The payoff is computed from (p_t, I_t) by a routine that reads no projection output;
 425 the optimal bundle x_t^* is a function of (p_t, I_t) alone and never references x_t or $\tilde{x}_t$. The share-fitted
 426 Cobb–Douglas demand of §3.1 is used only to log an upper bound on the reported distance. It is
 427 never passed to HiGHS as an incumbent solution, for which the solver exposes no interface.
- 428 **(2) Independence of the payoff optimum from revealed preference.** Raw (pre-repair) payoff does
 429 not differ between the 57 already-GARP-consistent traces and the 85 violating traces (0.7872 vs.
 430 0.7899; Welch $t = -0.07$, $p = 0.94$). No population-level link between GARP-consistency and
 431 payoff is detectable.
- 432 **(3) Trace-level mechanism check.** Five traces spanning the dose range (0.17 to 111.64) were
 433 inspected by hand, comparing the direction of the actual projection movement $(\tilde{x} - x)$ against the
 434 direction toward the exogenous optimum $(x^* - x)$ via cosine similarity; every quantity plotted in
 435 Figure 3 was recomputed by re-solving the projection MILP. Similarities were low (0.16–0.31) on the
 436 four traces where payoff improved after repair, and negative (−0.35) on the single trace where payoff
 437 worsened. On these five traces the sign of the cosine similarity matched the sign of the payoff change,
 438 and the cosine of the projection step against $(x^* - x)$ appears nowhere in the projection objective.
 439 This is evidence against the projection acting as a disguised optimization toward the payoff target.
- 440 **(4) Severity confound.** Larger-dose traces start from worse raw payoff (Pearson $r = -0.41$,
 441 $p < 0.0001$), and worse-starting traces improve more ($r = -0.41$ between raw payoff and Δ payoff).
 442 This is a ceiling-effect confound. The dose–response relationship survives it: the partial correlation
 443 of dose against Δ payoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$). The unconditional
 444 correlation is $r = 0.821$. In a joint OLS regression the dose coefficient is significant ($t = 11.43$,
 445 $p < 10^{-16}$) while raw payoff’s coefficient is not ($t = -1.36$, $p = 0.18$).
- 446 No part of the audit found a channel by which the projection could influence payoff other than through
 447 (p_t, I_t) .

448 B MILP formulation and implementation

- 449 **Formulation.** We use the multiplier-free ordinal characterization of Demuynck and Rehbeck [2023],
 450 under which every constraint is linear jointly in the perturbed bundles, a set of ordinal utility levels
 451 $u_t \in [0, 1]$, and binary comparison indicators $U_{t,v} \in \{0, 1\}$ for $t \neq v$. The formulation is solved as a
 452 single MILP, with no outer search over preference orderings. Afriat-multiplier parameterizations are
 453 bilinear once $\tilde{x}$ is a decision variable, and remain so under any fixed preference ordering, because a
 454 price-weighted multiplier term stays a product of two unknowns.
- 455 **Budget exhaustion.** Budget exhaustion is imposed as an equality, $p_t \cdot \tilde{x}_t = I_t$ for every observation,
 456 with income fixed at 100 throughout. The big- M constant in the ordering constraints is therefore
 457 fixed a priori: any $M > 100$ is valid, and it needs no tuning.
- 458 **Strict-preference margin.** The GARP-consistent set is not closed, so an unregularized projection
 459 can read a distance of exactly zero on violating data. A fixed strict-preference margin $\gamma > 0$, set to

460 $10^{-4} \cdot I = 10^{-2}$, converts an unattained infimum into an attained minimum. The reported distance is
 461 stable across $\gamma \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$, a sweep that runs downward from the operating
 462 point.

463 **Independent verification.** Every returned $\tilde{x}$ is verified by the Warshall-closure [Warshall, 1962]
 464 GARP check of §3.1. All 85 GARP-violating traces’ projections were re-verified GARP-consistent;
 465 MIP gap $\leq 8.1 \times 10^{-5}$ on every solve.

466 **Objective.** $\min \sum_t \sum_k w_{t,k} d_{t,k}$ subject to $\tilde{x}_{t,k} - x_{t,k} \leq d_{t,k}$ and $x_{t,k} - \tilde{x}_{t,k} \leq d_{t,k}$, with uniform
 467 weights $w = 1$.

468 B.1 Scope of the guarantee

469 The closest published analogue is a proved-optimal projection onto a convex monotone cone [Wang
 470 et al., 2026]. There, non-expansiveness licenses a guarantee that the edit lands weakly closer to
 471 ground truth, because the cone is fixed by an ordering supplied as input. The GARP-consistent set is
 472 a union of polyhedra rather than a single convex set, and a GARP repair must itself decide which
 473 revealed-preference comparisons to give up. That decision is absorbed into the binary comparison
 474 indicators $U_{t,v}$ rather than eliminated, so we claim no distance-minimization guarantee.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and §1 state exactly what is claimed (a measured dose–response relationship, an instrument-validity finding, and a replication of an independently published format effect) and what is not (novelty of repair, of projection, or of the sign of the effect); §6 states the design’s scope limits explicitly.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: §6 (Limitations) discusses CCEI’s noise and power at 1.5B, the single-run-per-condition design, the two-model-family scope, the single fixed payoff weighting, and the adverse prior from three prior negative results.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper’s contribution is a computational method (a MILP formulation for GARP-repair) building on a cited external representation theorem [Demuyne and Rehbeck, 2023]; it does not state or prove new theorems of its own.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: §4-5 specify the exact models, budget-set generator and seeding scheme, conditions, power calculation, retry protocol, and the full MILP formulation (§3) needed to reproduce the main results; the exogenous payoff is a closed-form function stated in full.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: An anonymized code and data package is included as supplementary material (prompts, seeds, the parser/validity rule, and raw per-trace logs for the main experiment, the null-operator control, the corrected-payoff robustness check, and the pilot), with a README giving the exact commands to regenerate every reported statistic from the included raw logs without re-running any model inference. Both LLMs used are publicly available open-weight models, so the full pipeline is also reproducible from scratch.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: §4 specifies model identities, all three conditions' exact framing, the budget-set generator and its rejection rule, seed scheme, replicate count and power calculation, and the discard/retry protocol.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Every reported comparison in §5 carries a p -value, and Table 1 and the dose-response statistics report means alongside sample sizes; CCEI comparisons additionally report standard deviations and, in §6, confidence-interval-based power limitations.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All inference was run locally on a single consumer CPU-class machine (no GPU, no commercial API calls), with the full $N = 30$ design measured at approximately 2.1 hours wall-clock (§4.1); every MILP projection solves in under 5 seconds with MIP gap $\leq 8.1 \times 10^{-5}$.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The study elicits only synthetic, non-sensitive budget-allocation choices from open-weight language models; it involves no human subjects, no personal data, and no scraped content.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: See the Broader impacts paragraph preceding §7: the paper’s own finding that 18% of repairs worsen the exogenous payoff is itself a caution against treating coherence-repair as a costless, universally beneficial deployment default.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper releases no model or dataset; both LLMs used are existing, publicly released open-weight models used unmodified as measurement subjects.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The two LLMs used are named with version in §4.1. Qwen2.5-1.5B-Instruct (Alibaba Cloud) is released under the Apache License 2.0. Llama-3.2-3B-Instruct (Meta Platforms, Inc.) is released under the Llama 3.2 Community License Agreement, a custom commercial license. Both are publicly available open-weight models used here for research purposes consistent with their license terms.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: This submission does not release a new dataset or model asset; the generated budget-set choice logs are experimental records, not a packaged new asset.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The study involves no crowdsourcing and no human subjects; all choices are elicited from language models.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The study involves no human subjects and required no IRB review.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: The two LLM agents under study are the paper's core experimental subject, not a writing aid: §4.1 names both models and their role as the choice-making agents whose revealed preferences are elicited, projected, and scored.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.